

Smoke Test

August 1, 2026

Calculus II Shared Math Library Smoke Test

This notebook verifies that the reusable `coursework_math` package can be imported and used inside the Berkeley Extension Calculus II project.

The smoke test covers:

- Python environment verification
- Shared package imports
- Equation plotting
- Shared image display
- Notebook export readiness

1. Verify the Python Environment

This section confirms that the notebook is running with the virtual environment created for the Calculus II project.

The expected interpreter path should end with:

`berkeley-ext-calculus-II\.venv\Scripts\python.exe`

Python executable: `c:\Users\britt\source\repos\math-through-code\berkeley-ext-calculus-II\.venv\Scripts\python.exe`

Working directory: `c:\Users\britt\source\repos\math-through-code\berkeley-ext-calculus-II\notebooks`

2. Verify the Shared Package

The `coursework_math` package is installed from the shared `math-library` project in editable mode.

An editable installation allows updates made to the shared library to become available to this course without copying the source files.

Package location: `C:\Users\britt\source\repos\math-through-code\math-library\src\coursework_math\__init__.py`

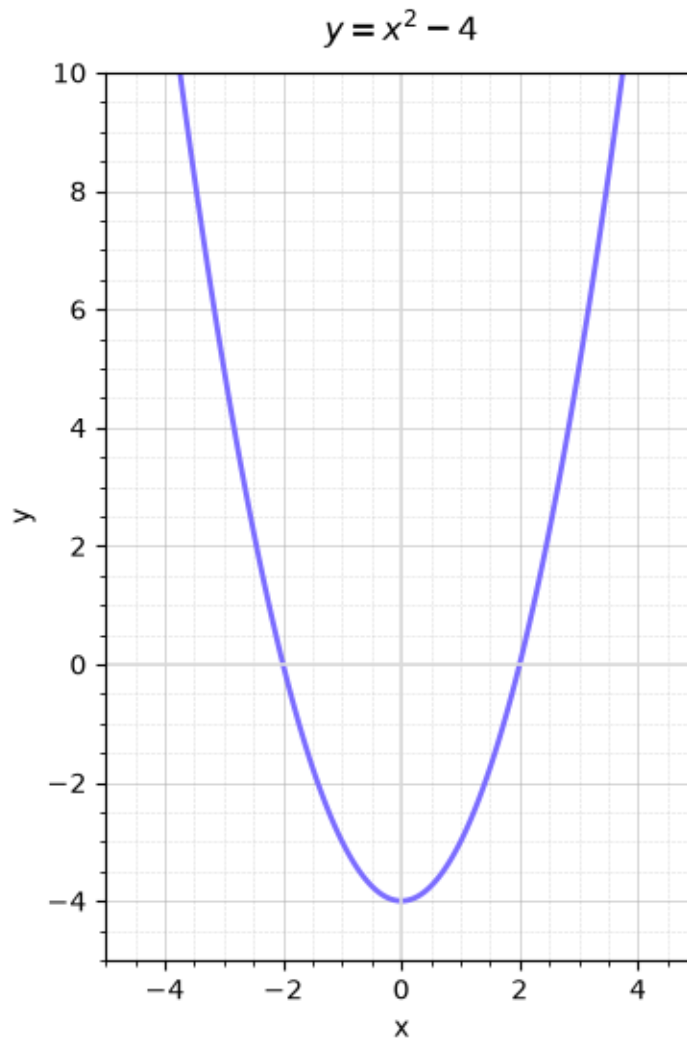
3. Test Equation Plotting

This example verifies that the shared plotting package can parse and graph an equation.

The graph below represents:

$$y = x^2 - 4$$

The function is a parabola shifted four units downward.



```
<Axes: title={'center': '$y = x^{2} - 4$'}, xlabel='x', ylabel='y'>
```

4. Locate a Shared Knowledge Image

Shared images are stored in the math library rather than duplicated inside individual course repositories.

This section builds the image path relative to the Calculus II project and confirms that the file exists before displaying it.

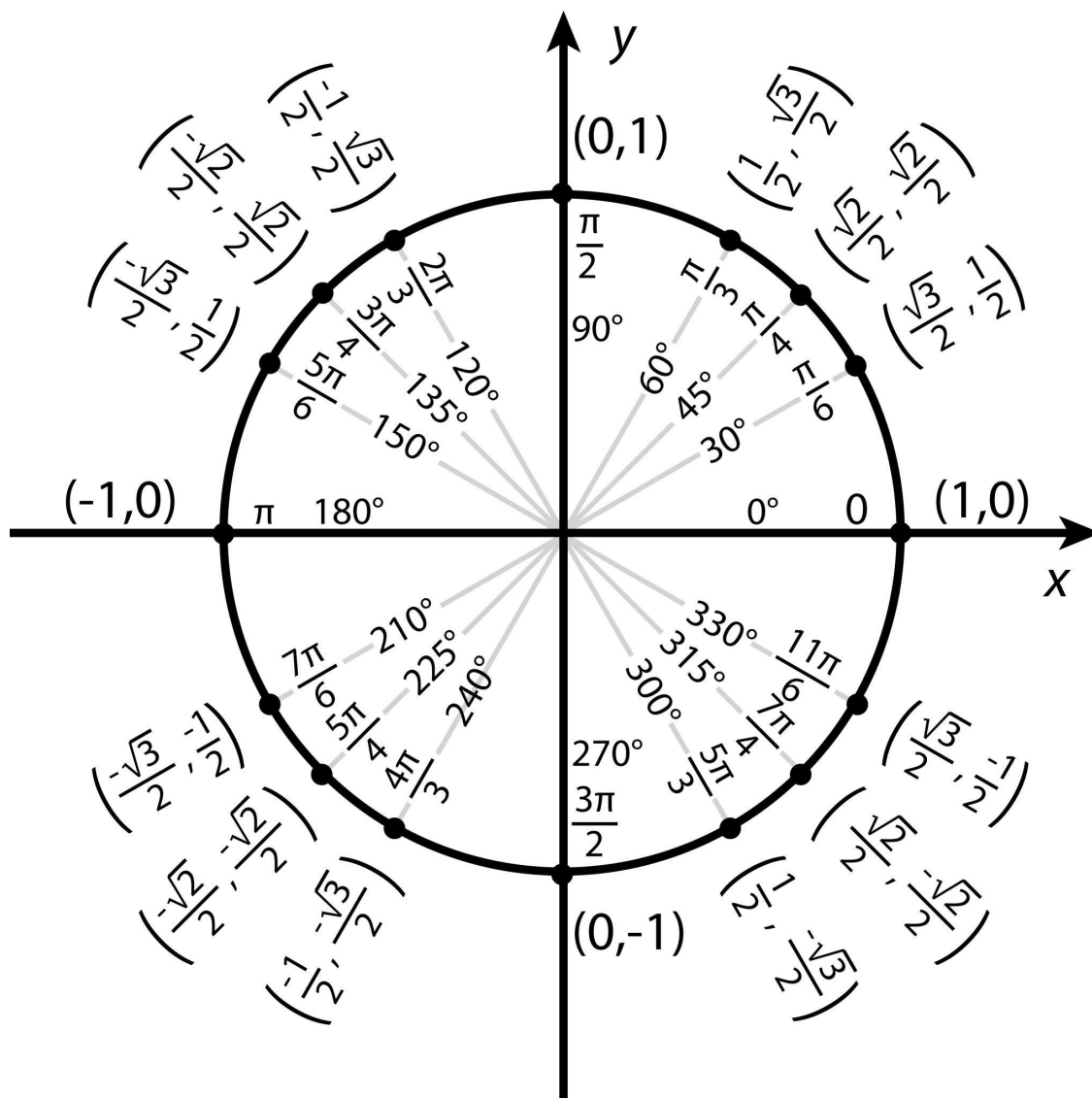
Image path: c:\Users\britt\source\repos\math-through-code\math-library\knowledge\images\unit_circle.jpg
Image exists: True

5. Test Shared Image Display

The unit-circle image below is loaded through the shared image utility.

This verifies that:

1. The image path is valid.
2. Pillow can open the image.
3. The image can be displayed inside a Jupyter notebook.



6. Notebook Export Command

The shared library provides the `math-notebook-pdf` command for exporting notebooks to PDF.

From the Calculus II project terminal, this notebook can be exported with:

```
math-notebook-pdf notebooks/smoke-test.ipynb
```

Results

7. Smoke-Test Results

Test	Expected Result	Status
Calculus II virtual environment	<code>.venv</code> interpreter is active	
<code>coursework_math</code> import	Package path is displayed	
Plotting import	Equation graph is rendered	
Image path	File exists	
Image utility	Unit-circle image is displayed	
Export command	<code>math-notebook-pdf --help</code> works	

Conclusion

The Calculus II project can successfully use the shared `coursework_math` package.

Future notebooks can import reusable plotting, image, and notebook-export utilities without copying implementation files into the course repository.

ChatGPT was used for Markdown and L^AT_EX formatting assistance. All mathematical reasoning, derivations, and conclusions are my own work and were independently checked for correctness.